

Cell Biology (BT251) Course Syllabus

Course Description

This course will provide an illustrated, up-to-date introduction to cell biology concepts, processes, and applications.

Course Textbook

Becker's World of the Cell. Jeff Hardin, Gregory Paul Bertoni and Lewis J. Kleinsmith 9th Edition.

Course Content

1. A Preview of Cell Biology
2. The Chemistry of the Cell
3. The Macromolecules of the Cell
4. Cells and Organelles
7. Membranes: Their Structure, Function, and Chemistry
8. Transport Across Membranes: Overcoming the Permeability Barrier
12. The Endomembrane System
13. Cytoskeletal Systems
14. Cellular Movement: Motility and Contractility
15. Beyond the Cell: Cell Adhesions, Cell Junctions, and Extracellular Structures
22. Signal Transduction Mechanisms: I. Electrical and Synaptic Signaling in Neurons
23. Signal Transduction Mechanisms: II. Messengers and Receptors
24. The Cell Cycle and Mitosis
26. Cancer Cells

Grades and exams information

- The course final grade will be based on the First (30%), Second (30%) and Final (40%) exams.

E-learning

The course is partially supported by E-learning, a web-based program that enables you to access PowerPoint slides and course announcements.